Saransh Agrawal

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ML Engineer with production experience in LLM infrastructure, distributed training, and backend systems. Skilled in inference optimization, RAG pipelines and ML platform development. Published researcher (ACL/EACL) with a focus on model alignment, scaling e�ciency, and mechanistic interpretability.

Education

Texas A&M University *Master of Science in Data Science: (GPA: 3.7/4.0)* — *College Station, TX, **May 2025***

Courses: Parallel Computing, Database Mgmt., Natural Language Processing, Reinforcement Learning, Data Stream & Algo.

Manipal Institute of Technology *Bachelors in Electronics and Instrumentation Engineering* — ***Aug 2021***

Courses: Data Structures & Algorithms, Microprocessors, Digital Image Processing, Embedded Systems, Digital System Design

Technical Skills

AI/ML Systems: PyTorch, DeepSpeed, Triton, vLLM, llama.cpp, RAG Pipelines, Vector DBs (Qdrant)
Languages: Python, Rust, Go, JavaScript, C++, SQL
Infrastructure: AWS (SageMaker, Bedrock, EKS, Lambda, EC2, S3), Kafka, Docker, Kubernetes, Terraform, HPC
Certificates: AWS Solutions Architect - Professional

Work Experience

Independent Open Source Contributor & Researcher — **May 2025 - Present**

- **NVIDIA/kvpress (Open Source)**: Developed a Triton-accelerated KV-cache merging kernel, achieving a 175x speedup (w.r.t PyTorch) by optimizing memory access patterns and eliminating redundant VRAM-SRAM transfers.
- **Saptarshi Software (Volunteer)**: Architecting a high-concurrency RAG orchestrator in Go (Echo/Redis); built a modular Ollama inference pipeline and IAM/API Gateway for multi-tenant agentic workflows.
- **End-to-End RAG Optimization**: Engineered an agentic RAG pipeline achieving <900ms Time-to-First-Token (TTFT); optimized the Go (Echo) orchestrator and Qdrant hybrid search.
- **Semantic Search Engine**: Engineered a ranking engine for 80k+ ACL papers using Neo4j and Sentence-Transformers; implemented a hybrid PageRank/embedding-based similarity algorithm.

Research Assistant | *FLAIR Lab* (khhuang.me/group.html) — **Aug 2024 – May 2025**

NLP Research Lab at Texas A&M University under Prof. Kuan-Hao Huang — *College Station, TX*

- **LLM Alignment/Unlearning**: Won 2nd Prize (1B) and highest score (8B) at SemEval-2025; published at ACL (SemEval) on mechanistic interpretability and constrained knowledge isolation.
- **Distributed Training Infra**: Optimized multi-node 27B model fine-tuning on A100 clusters; tuned NCCL/Infiniband parameters to achieve 92% scaling e�ciency using DeepSpeed ZeRO-3.
- **Inference Optimization**: Developed a vision-language KV-cache merging algorithm (CaM) for LLaVA, reducing VRAM by 21% while maintaining context retrieval accuracy.
- **Founding Research Member**: Established lab research on mechanistic interpretability; collaborated with NTU Taiwan on DPO-based preference vectors and built the distributed LLM evaluation framework.

Software Development Engineer | *Viewzen Labs* (viewzenlabs.com) — **July 2021 – July 2023**

Enterprise MLaaS and data platform startup serving finance, healthcare, and government clients.

- **Multi-tenant MLaaS Platform**: Architected a scalable platform using Go, Python and Kafka to orchestrate experiment tracking and low-latency inference (Sklearn Models) for 50+ enterprise clients.
- **High-Throughput ETL Engine**: Engineered fault-tolerant microservices using a C++/Python FFI, achieving a 15x throughput increase (w.r.t. Node.js) for hierarchical data migrations.
- **Production ML Deployment**: Trained and deployed automated XGBoost risk models for maternal healthcare, achieving an 0.85 F1-score; managed the full MLOps Lifecycle.

Machine Learning Engineer Intern | *Centre for Digital Financial Inclusion* (cdfi.in) — **Feb 2021 – June 2021**

Non-profit developing tech solutions for e-Governance and fintech.

- **Speech-to-SQL Pipeline**: Developed a voice-activated dashboard using T5 and schema-guided decoding, achieving 90% SQL validity with sub-4s end-to-end latency.
- **ASR Model Fine-tuning**: Reduced Word Error Rate (WER) to 17% by fine-tuning DeepSpeech models on regional Indian accent corpora for e-governance applications.

Publications

[1] **Saransh Agrawal** and Kuan-Hao Huang. Selective Amnesia – Constrained Unlearning for Large Language Models via Knowledge Isolation. In *Proceedings of the 19th International Workshop on Semantic Evaluation (SemEval-2025) at ACL*.

[2] Ren-Wei Liang, Chin-Ting Hsu, Chan-Hung Yu, **Saransh Agrawal**, Shih-Cheng Huang, Shang-Tse Chen, Kuan-Hao Huang, and Shao-Hua Sun. Adaptive Helpfulness-Harmlessness Alignment with Preference Vectors. In *European Chapter of the Association for Computational Linguistics (EACL) 2026*